

Sydney Girls High School

2003
TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics

Extension 2

This is a trial paper ONLY.
It does not necessarily
reflect the format or the
contents of the 2003 HSC
Examination Paper in this
subject.

General Instructions

- ♦ Reading Time – 5 mins
- ♦ Working Time – 3 hours
- ♦ Attempt ALL questions
- ♦ ALL questions are of equal value
- ♦ All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- ♦ Standard integrals are supplied
- ♦ Board-approved calculators may be used.
- ♦ Diagrams are not to scale
- ♦ Each question attempted should be started on a new sheet. Write on one side of the paper only.

Question 1.

a) Evaluate. i) $\int_0^2 \frac{x}{x^2+4} dx$

ii) $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \sin 2x \cdot \cos x \cdot dx$

iii) $\int_1^2 x^2 \log_e x \cdot dx$

b) Let n be a positive integer, and let $I_n = \int_1^2 (\log_e x)^n \cdot dx$.

prove that $I_n = 2(\log_e 2)^n - n \cdot I_{n-1}$ and hence evaluate

$\int_1^2 (\log_e x)^3 \cdot dx$ as a polynomial in $\log_e 2$

Question 2.

a) i) Find $\sqrt{-3-4i}$

ii) Solve the equation $x^2 - 3x + 3 + i = 0$ over the complex field

b) i) Show that there are two complex numbers z such that

$|z - 2 - i| = 1$ and $\arg z = \frac{\pi}{4}$,

ii) Find the moduli of the two values of z found in part i)

c) A point P representing the complex number z moves in the Argand Diagram so this it lies in the region defined by:

$|z - 1| \leq |z - i|$ and $|z - 2 - 2i| \leq 1$

i) Indicate on a sketch, the region within which P lies

ii) If P describes the boundary of the region, find

α) the value of $|z|$ when $\arg z$ has its smallest value

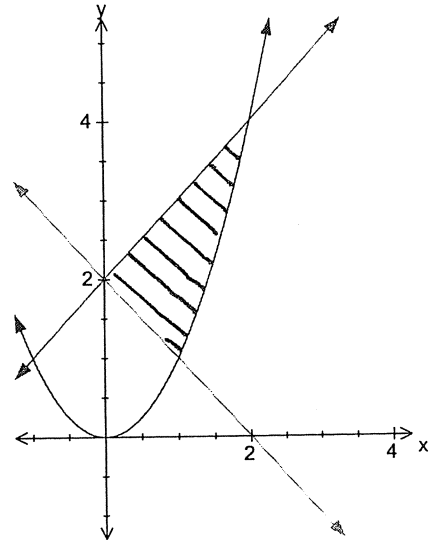
β) the values of z in the form $a + ib$ when $\arg (z-1) = \frac{\pi}{4}$

Question 3:

- a) The adjacent diagram shows the area enclosed by $y = 2 - x$, $y = 2 + x$ and $y = x^2$.

The area is to be rotated about the Y axis.

- Find the shaded area
- Find the volume that is formed when it is rotated



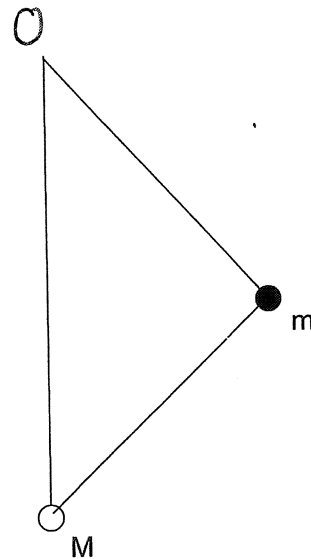
- b) A satellite moves in a circular orbit of radius 8000km, making 12 revolutions per day. Find:

- the velocity of the satellite
- the centripetal force acting on the satellite if the mass of that satellite is 500kg.

- c) A particle of mass m is attached to a fixed point O by a string of length one metre, and by another string of the same length to a small ring of mass M which can slide on a smooth vertical wire underneath O . If m describes a horizontal circle with constant angular velocity ω , prove that

its depth below O is $\left(\frac{m + 2M}{m\omega^2} \right)g$,

where g is acceleration due to gravity



Questions 4:

- a) (i) Sketch $\frac{x^2}{4} + \frac{y^2}{9} = 2$, indicating the centre, foci and directrices

- (ii) If $P(2\sqrt{2}\cos\theta, 3\sqrt{3}\sin\theta)$ lies on the ellipse find

α) The equation of the normal at P

β) The value of θ to the nearest degree if the normal passes through the point $(-2\sqrt{2}, 0)$

b) $P\left(3p, \frac{3}{p}\right)$ and $Q\left(3q, \frac{3}{q}\right)$ lie on the hyperbola $xy = 9$.

(i) Find the equation of the tangent at P

(ii) Find the point of intersection T, of the tangents at P and Q.

(iii) If the chord of contact from T passes through the point (0,2) find the locus of T.

Question 5.

a) Given $f(x) = \frac{7x}{(x^2+3)(x+2)}$

i) Express $f(x)$ as a sum of partial fractions

ii) Evaluate $\int_0^3 f(x).dx$

b) Without the use of calculus, sketch the following curves

i) $y = \frac{x(x-2)}{x-1}$

ii) $y = \frac{x(x-1)}{x-2}$

c) Consider $y = \frac{x^3}{(x-1)^2}$

i) Determine the asymptotes

ii) Determine the stationary points

iii) Sketch the curve showing any important features

Question 6.

a) (i) Show that if a polynomial $P(x)$ has a root b of multiplicity m , the the polynomial $P'(x)$ has the root b with multiplicity $m-1$

(ii) Given that $Q(x) = x^4 - 5x^3 + 4x^2 + 3x + 9$ has a zero of multiplicity 2, solve the equation $Q(x) = 0$ over the complex field

b) If α, β and γ are the roots of $3x^3 + 4x^2 + 5x + 1 = 0$, find the value of $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\gamma^2\alpha^2}$

c) i) If $x^2 - 3x + 4 = 0$, show that $x^4 = 3x - 20$

ii) Hence or otherwise, find the equation with roots α^4 and β^4 if the roots of $x^2 - 3x + 4 = 0$ are α and β .

Question 7:

a) Show that the volume of the largest cylinder that can be cut from

a solid sphere of radius r cm is $\frac{4\pi r^3}{3\sqrt{3}}$ cm³

b) (i) Find the five roots of $z^5 = 1$ and write them in mod-arg form.

(ii) Show that when these five roots are plotted on an Argand

Diagram, they form the vertices of a regular pentagon

of area $\frac{5}{2} \sin \frac{2\pi}{5}$

(iii) Factorise $z^5 - 1$ over the real field

(iv) Deduce that $\cos \frac{2\pi}{5}$ is a root of the equation $4x^2 + 2x - 1 = 0$

and hence find the exact value of $\cos \frac{2\pi}{5}$.

Question 8:

a) A particle of mass m falls from rest at a height h above the

earth's surface, against a resistance kv per unit mass when its speed is v ; k being a positive constant.

(i) Show that its equation of motion may be written in the form

$$v \frac{dv}{dx} = g - kv$$

(ii) If the particle reaches the surface of the earth with speed V , show that

$$\log_e \left(1 - \frac{kV}{g} \right) + \frac{kV}{g} + \frac{k^2 h}{g} = 0$$

b) (i) A particle P is projected from a point O on horizontal ground,

with speed V at an angle $\theta = \tan^{-1} \left(\frac{1}{3} \right)$. The particle passes through

the point with co-ordinates $\left(3a, \frac{3a}{4} \right)$. Show that $V^2 = 20ga$.

(ii) A particle Q is projected from the same point O at the instant when

P reaches its maximum height. It strikes the ground at the same place and time

as P strikes the ground. Show that the speed of projection of Q is $\sqrt{\frac{145ga}{2}}$

and find the tangent of the angle of projection.

Question 1

$$a) \int_0^2 \frac{x}{x^2+4} dx = \frac{1}{2} \int_0^2 \frac{2x}{x^2+4} dx$$

$$= \frac{1}{2} [\log_e (x^2+4)]_0^2$$

$$= \frac{1}{2} [\log_e 8 - \log_e 4]$$

$$= \underline{\underline{\frac{1}{2} \log_e 2}} \quad (3)$$

$$ii) \int_{-\pi}^{\pi} \sin 2x \cos x dx = \underline{\underline{0}}$$

as fn odd [sin 2x odd, cos x even]

$$iii) \int_1^2 x^2 \log_e x dx$$

$$\text{let } u = \log_e x, \quad v = x^3, \quad u = \frac{1}{x}, \quad v = \frac{x^3}{3}$$

$$I = \frac{x^3}{3} \log_e x - \int \frac{1}{x} \cdot \frac{x^3}{3} dx$$

$$= \frac{x^3}{3} \log_e x - \frac{1}{3} \int x^2 dx$$

$$= \underline{\underline{\frac{x^3}{3} \log_e x - \frac{x^3}{9}}} \quad 3$$

$$\int_1^2 x^2 \log_e x dx = \left[\frac{x^3}{3} \log_e x - \frac{x^3}{9} \right]_1^2$$

$$= \left[\frac{8}{3} \log_e 2 - \frac{8}{9} \right] - \left[\frac{1}{3} \log_e 1 - \frac{1}{9} \right]$$

$$= \underline{\underline{\frac{8}{3} \log_e 2 - \frac{7}{9}}} \quad (4)$$

$$b) I_n = \int_1^2 (\log_e x)^n dx$$

$$\text{let } u = (\log_e x)^n, \quad v = 1, \quad u = \frac{n}{x} (\log_e x)^{n-1}, \quad v = x$$

$$\therefore I_n = [x (\log_e x)^n]_1^2 - \int_1^2 n (\log_e x)^{n-1} dx$$

$$I_n = 2 (\log_e 2)^n - n I_{n-1} \quad (3)$$

$$I_3 = 2 (\log_e 2)^3 - 3 I_2$$

$$= 2 (\ln 2)^3 - 3 [(2 \ln 2)^2 - 2 I_1] \quad (2)$$

$$= 2 (\ln 2)^3 - 6 (\ln 2)^2 + 6 [2 \ln 2 - I_0]$$

$$= \underline{\underline{2 (\ln 2)^3 - 6 (\ln 2)^2 + 12 \ln 2 - 6}}$$

Question 2

a) i) let $a + bi = \sqrt{-3 - 4i}$

$$a^2 + b^2 + 2abi = -3 - 4i$$

$$a^2 - b^2 = -3 \quad (1), \quad 2ab = -4$$

$$(a^2 + b^2)^2 = (a^2 - b^2)^2 + 4a^2b^2$$

$$= 9 + 16$$

$$a^2 + b^2 = 5 \quad (2) \quad (a^2 + b^2 > 0)$$

$$(1) + (2) \quad 2a^2 = 2$$

$$a = \pm 1 \quad b = \mp 2$$

$$\therefore \sqrt{-3 - 4i} = \underline{\underline{\pm (1 - 2i)}} \quad (3)$$

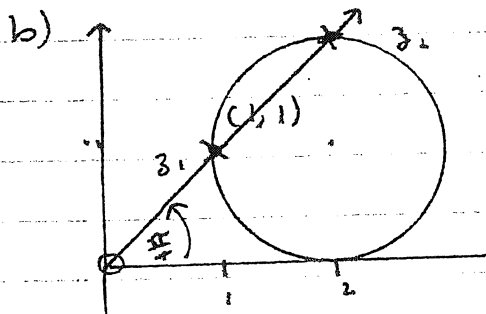
ii) $z^2 - 3z + 3 + i = 0$

$$z = 3 \pm \frac{9 - \sqrt{4(1)(-3+i)}}{2}$$

$$= \frac{3 + \sqrt{-3 - 4i}}{2}, \frac{3 - \sqrt{-3 - 4i}}{2}$$

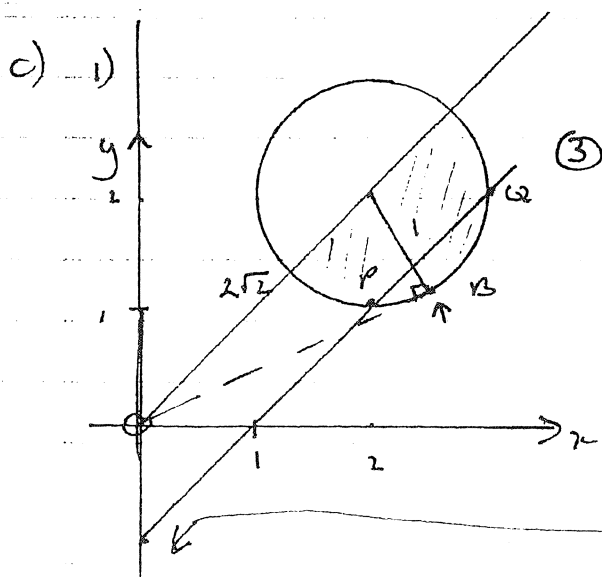
$$= \frac{3 + (1 - 2i)}{2}, \frac{3 - 1 - 2i}{2}$$

$$= \underline{\underline{(2 - i)}}, \underline{\underline{(1 + i)}} \quad (2)$$



i) from diagram $z_1(1, 1)$, $z_2(2, 2)$
(or solve algebraically) (1)

$$ii) |z_1| = \sqrt{1^2 + 1^2} = \frac{\sqrt{2}}{1} \quad |z_2| = \sqrt{2^2 + 2^2} = \frac{2\sqrt{2}}{1}$$



ii) $\arg z$ min at B, $OB = \sqrt{8}$
(2) $= \sqrt{7}$

B) $\arg(z - 1) = \frac{\pi}{4}$ is
the line $y = x - 1$
cuts circle at P, C

$$\text{Solving } (x - 2)^2 + (y - 2)^2 = 1$$

$$y = x - 1 \quad \text{or}$$

$$(x - 2)^2 + (x - 3)^2 = 1 \quad \text{From Diag.}$$

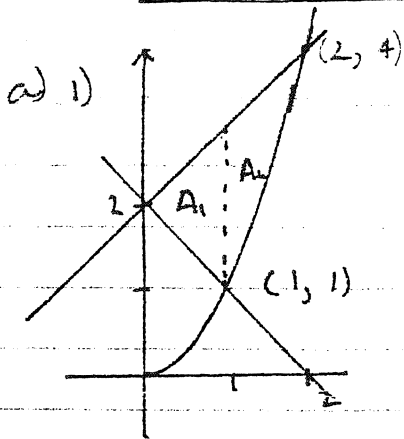
$$x^2 - 4x + 4 + x^2 - 6x + 9 - 1 = 0 \quad | \quad (x - 3)(x - 2) = 0$$

$$2x^2 - 10x + 12 = 0$$

$$x^2 - 5x + 6 = 0$$

$$\left. \begin{array}{l} x = 2 \\ y = 1 \end{array} \right\} \quad , \quad \left. \begin{array}{l} x = 3 \\ y = 2 \end{array} \right\}$$

Question 3



$$\begin{aligned}
 A_1 &= \int_0^1 [(2+x) - (2-x^2)] dx \\
 &= \int_0^1 2x dx \\
 &= [x^2]_0^1 \\
 &= 1 \text{ unit}^2
 \end{aligned}$$

$$\begin{aligned}
 A_2 &= \int_1^2 (2+x - x^2) dx \\
 &= \left[2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_1^2 \\
 &= \left(4 + 2 - \frac{8}{3} \right) - \left(2 + \frac{1}{2} - \frac{1}{3} \right) \\
 &= \frac{1}{6} \text{ unit}^2
 \end{aligned}$$

$$\therefore A = 2 \frac{1}{6} \text{ unit}^2 \quad (3)$$

$$\begin{aligned}
 \text{ii) } V_{\text{shell 1}} &= \pi [R^2 - r^2] h \quad (\text{shell 1}) \\
 &= \pi [(x+\Delta x)^2 - x^2] [2x] \\
 &= \pi [x^2 + 2x\Delta x + (\Delta x)^2 - x^2] 2x \\
 &= \pi (4x^2) \Delta x \quad [(\Delta x)^2 \text{ v. small}]
 \end{aligned}$$

$$\begin{aligned}
 V_{\text{solid 1}} &= 4\pi \int_0^1 x^2 dx \\
 &= \frac{4\pi}{3} [x^3]_0^1 \\
 &= \frac{4\pi}{3} \text{ unit}^3
 \end{aligned}$$

$$\begin{aligned}
 V_{\text{shell 2}} &= \pi [R^2 - r^2] h \quad (\text{shell 2}) \\
 &= \pi [(x+\Delta x)^2 - x^2] (2+x-x^2) \\
 &= \pi [x^2 + 2x\Delta x + (\Delta x)^2 - x^2] (2+x-x^2) \\
 &= \pi [2x\Delta x] (2+x-x^2) \\
 &= 2\pi [2x + x^2 - x^3] \Delta x
 \end{aligned}$$

$$\begin{aligned}
 V_{\text{solid 2}} &= 2\pi \int_1^2 (2x + x^2 - x^3) dx \\
 &= 2\pi \left[x^2 + \frac{x^3}{3} - \frac{x^4}{4} \right]_1^2 \\
 &= 2\pi \left[\left(4 + \frac{8}{3} - \frac{16}{4} \right) - \left(1 + \frac{1}{3} - \frac{1}{4} \right) \right] \\
 &= \frac{19\pi}{6} \text{ unit}^3
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Volume} &= \frac{4\pi}{3} + \frac{19\pi}{6} \\
 &= \frac{9\pi}{2} \text{ unit}^3 \quad (4)
 \end{aligned}$$

Quest 3 (continued)

b) i) $r = 8000 \text{ km} = 8000000 \text{ m}$

$$\omega = \frac{1 \cancel{\text{K}} \times 2\pi}{24 \times 60 \times 60} \text{ rad/s}$$

$$= \frac{\pi}{3600} \text{ rad/s}$$

$$v = r\omega$$

$$= 8000000 \times \frac{\pi}{3600}$$

$$= \frac{20000\pi}{9} \text{ ms}^{-1}$$

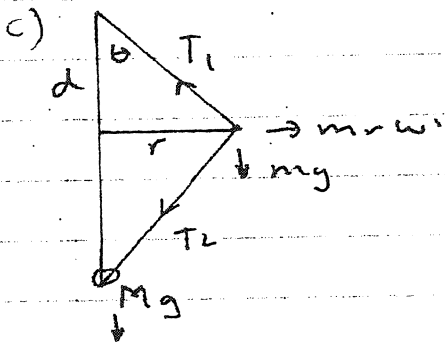
$$\textcircled{2} (\div 6981.3) \text{ ms}^{-1}$$

ii) $F = m r \omega^2$

$$= 500 \times 8000000 \times \frac{\pi^2}{(3600)^2}$$

$$\div \underline{3046 \text{ N}}$$

$\textcircled{2}$



At M, $T_2 \cos \theta = Mg \Rightarrow T_2 = \frac{Mg}{\cos \theta}$

at m, $T_1 \cos \theta = T_2 \cos \theta + mg$ (vert)

ie $T_1 \cos \theta = Mg + mg \Rightarrow T_1 = \frac{Mg + mg}{\cos \theta}$

$T_1 \sin \theta + T_2 \sin \theta = m r \omega^2$ (hor)

but $r = d \sin \theta$

$$T_1 + T_2 = m \omega^2 d$$

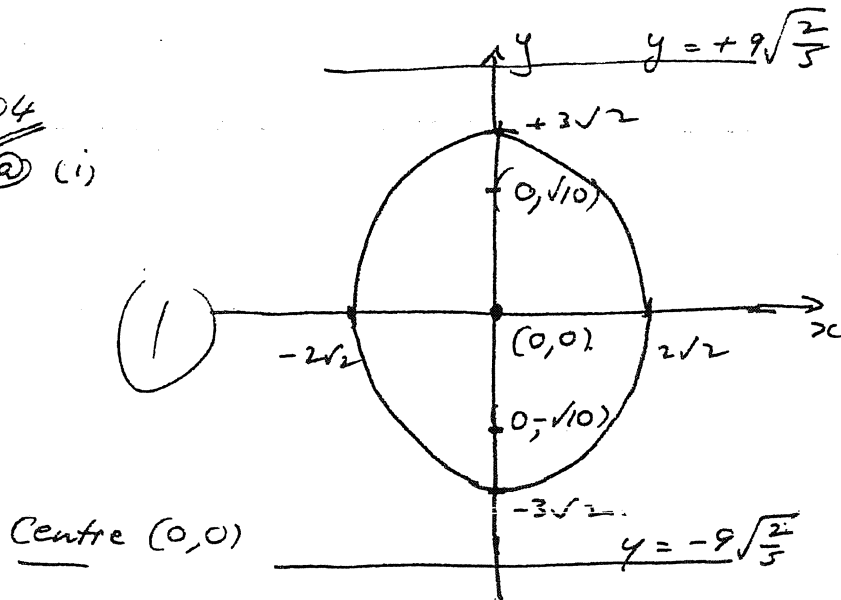
$$\frac{Mg + mg}{\cos \theta} + \frac{Mg}{\cos \theta} = m \omega^2 d$$

$$\frac{2Mg + mg}{m \omega^2} = d \cos \theta$$

but $d = \frac{2M + m}{m \omega^2} g$

$$d = \left(\frac{2M + m}{m \omega^2} \right) g \quad \textcircled{4}$$

Q4
Q (i)



Centre (0,0)

Foci (0, $\pm be$)

(1) Foci (0, $\pm 3\sqrt{2} \cdot \frac{\sqrt{5}}{3}$)

Foci (0, $\pm \sqrt{10}$)

(1) Directrices $y = \pm \frac{b}{e} = \pm \frac{3\sqrt{2}}{\frac{\sqrt{5}}{3}} = \pm \frac{9\sqrt{2}}{\sqrt{5}} = \pm 9\sqrt{\frac{2}{5}}$
 $y = \pm 9\sqrt{\frac{2}{5}}$

$$\frac{x^2}{4} + \frac{y^2}{9} = 2$$

$$\frac{x^2}{8} + \frac{y^2}{18} = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\therefore a = 2\sqrt{2}$$

$$b = 3\sqrt{2}$$

$$a^2 = b^2(1 - e^2)$$

$$e^2 = \frac{b^2 - a^2}{b^2}$$

$$\therefore e = \frac{18 - 8}{18} = \frac{10}{18} = \frac{5}{9}$$

$$e = \frac{\sqrt{5}}{3}$$

Q4
a) (ii) 2)

C Using $P(2\sqrt{2}\cos\theta, 3\sqrt{3}\sin\theta)$

$$y = 3\sqrt{3}\sin\theta$$

$$x = 2\sqrt{2}\cos\theta$$

$$\frac{dy}{d\theta} = 3\sqrt{3}\cos\theta$$

$$\frac{dx}{d\theta} = -2\sqrt{2}\sin\theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = 3\sqrt{3}\cos\theta \times \frac{1}{-2\sqrt{2}\sin\theta}$$

$$\text{Grad of tangent} : \frac{3\sqrt{3}\cos\theta}{-2\sqrt{2}\sin\theta}$$

$$\text{Grad of normal} : \frac{2\sqrt{2}\sin\theta}{3\sqrt{3}\cos\theta}$$

Eqa of normal :

$$y - 3\sqrt{3}\sin\theta = \frac{2\sqrt{2}\sin\theta}{3\sqrt{3}\cos\theta} (x - 2\sqrt{2}\cos\theta)$$

$$y - 3\sqrt{3}\sin\theta = \frac{2\sqrt{2}\sin\theta}{3\sqrt{3}\cos\theta} x - \frac{8\sin\theta}{3\sqrt{3}}$$

$$y = \frac{2\sqrt{2}\sin\theta}{3\sqrt{3}\cos\theta} x + 3\sqrt{3}\sin\theta - \frac{8\sqrt{3}\sin\theta}{9}$$

$$y = \frac{2\sqrt{2}\sin\theta}{3\sqrt{3}\cos\theta} x + \frac{19\sqrt{3}\sin\theta}{9}$$

b)

Passes thru $(-2\sqrt{2}, 0)$.

$$\therefore 0 = \frac{-8\sin\theta}{3\sqrt{3}\cos\theta} + \frac{19\sqrt{3}\sin\theta}{9}$$

$$\frac{8\sin\theta}{3\sqrt{3}\cos\theta} = \frac{19\sqrt{3}\sin\theta}{9}$$

$$\Rightarrow 8 = \frac{19 \times 9 \cos\theta}{9}$$

$$\cos\theta = \frac{8}{19}$$

$$\underline{\underline{\theta = 65^\circ}}$$

Q4

(a) (ii) & P(2/2 cos θ, 3/3 sin θ).

A) Using $\frac{x^2}{4} + \frac{y^2}{9} = 2$.

$$(1) \quad \frac{2x}{4} + \frac{2y}{9} \cdot \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{9x}{4y}.$$

$$\text{At } P(2/2 \cos \theta, 3/3 \sin \theta) \quad \frac{dy}{dx} = \frac{-18/2 \cos \theta}{12/3 \sin \theta} = \frac{-3/2 \cos \theta}{2/3 \sin \theta}$$

$$(1) \quad \text{Grad of normal: } \frac{+2/3 \sin \theta}{3/2 \cos \theta}.$$

$$\text{Eqn of normal. } y - 3/3 \sin \theta = \frac{2/3 \sin \theta}{3/2 \cos \theta} (x - 2/2 \cos \theta).$$

$$y - 3/3 \sin \theta = \frac{2/3 \sin \theta}{3/2 \cos \theta} x - \frac{4}{3} \sqrt{3} \sin \theta.$$

(2)

$$y = \frac{2/3 \sin \theta}{3/2 \cos \theta} x + \frac{5}{3} \sqrt{3} \sin \theta.$$

B) Passing thru (-2/2, 0)

(1)

$$0 = \frac{2/3 \sin \theta}{3/2 \cos \theta} x - 2/2 + \frac{5}{3} \sqrt{3} \sin \theta$$

$$+10\sqrt{3} \sin \theta = \frac{2/3 \sin \theta}{\cos \theta}$$

$$5 \cos \theta = 1$$

(1)

$$\cos \theta = \frac{1}{5}$$

$$\theta = 78^\circ$$

Q4

(ii) $\propto$ $\frac{x^2}{4} + \frac{y^2}{9} = 2 \Rightarrow \frac{dy}{dx} = -\frac{9x}{4y}$

B Using: $P(2\sqrt{2}\cos\theta, 3\sqrt{2}\sin\theta)$.

Gradient: $\frac{dy}{dx} = -\frac{9 \cdot 2\sqrt{2}\cos\theta}{4 \cdot 3\sqrt{2}\sin\theta} = -\frac{3\cos\theta}{2\sin\theta}$.

Gradient of Normal.

$$\frac{dy}{dx} = \frac{2\sin\theta}{3\cos\theta}.$$

Equation of normal.

$$y - 3\sqrt{2}\sin\theta = \frac{2\sin\theta}{3\cos\theta} (x - 2\sqrt{2}\cos\theta)$$

$$y - 3\sqrt{2}\sin\theta = \frac{2\sin\theta}{3\cos\theta} x - \frac{4\sqrt{2}\sin\theta}{3}$$

$$\therefore y = \frac{2\sin\theta}{3\cos\theta} x - \frac{4\sqrt{2}}{3} + 3\sqrt{2}\sin\theta$$

$$y = \frac{2\sin\theta}{3\cos\theta} x + \frac{5\sqrt{2}\sin\theta}{3}$$

Passes thru $(-2\sqrt{2}, 0)$

$$0 = \frac{2\sin\theta}{3\cos\theta} \cdot -2\sqrt{2} + \frac{5\sqrt{2}\sin\theta}{3}$$

$$\therefore \frac{4\sqrt{2}\sin\theta}{3\cos\theta} = \frac{5\sqrt{2}\sin\theta}{3}$$

$$\cos\theta = \frac{4}{5}$$

$$\underline{\underline{\theta = 37^\circ}}$$

Q4 (b) $P(3p, \frac{3}{p})$ and $Q(3q, \frac{3}{q})$ lie on $xy = 9$.

(i) $y = 9x^{-1}$

$$\frac{dy}{dx} = \frac{-9}{x^2}$$

But $x = 3p$ $\frac{dy}{dx} = \frac{-9}{9p^2} = \frac{-1}{p^2}$

(2) Eqn of tangent $y - \frac{3}{p} = -\frac{1}{p^2}(x - 3p)$
 $p^2y - 3p = -x + 3p$
 $x + p^2y = 6p$ *

(iii) Tangent at P $x + p^2y = 6p$ — (1)

" " Q $x + q^2y = 6q$ — (2)

$$y(p^2 - q^2) = 6(p - q)$$

$$\therefore y = \frac{6}{p + q}$$

$$q^2x + p^2q^2y = 6pq^2$$

$$p^2x + p^2q^2y = 6qp^2$$

$$x(q^2 - p^2) = 6pq(q - p)$$

(2)

$$x = \frac{6pq}{p + q}, y = \frac{6}{p + q}$$

$$T\left(\frac{6pq}{p + q}, \frac{6}{p + q}\right)$$

(iii) Chord of contact

(1) $xy_0 + yx_0 = 18$

$$-2\left(\frac{6pq}{p + q}\right) = 18$$

$$x\left(\frac{6}{p + q}\right) + y\left(\frac{6p}{p + q}\right) = 18$$

$$\frac{6pq}{p + q} = 9$$

passes thru (0, 2)

But $x = \frac{6pq}{p + q}$

$\therefore$ Locus is

(1) $x = 9$

Q5 (i) $f(x) = \frac{7x}{(x^2+3)(x+2)}$

$$\frac{7x}{(x^2+3)(x+2)} = \frac{Ax+B}{x^2+3} + \frac{C}{x+2}$$

$$\Rightarrow (Ax+B)(x+2) + C(x^2+3) = 7x$$

$$x=-2 \Rightarrow 7C = -14 \Rightarrow C = -2.$$

$$\text{Equate } x^2 \Rightarrow A+C=0 \therefore A=+2.$$

$$x=0 \quad 2B+3C=0$$

$$2B-6=0 \Rightarrow B=3.$$

2

$$\therefore f(x) = \frac{2x+3}{x^2+3} - \frac{2}{x+2}.$$

(ii) $I = \int_0^3 \frac{7x \cdot dx}{(x^2+3)(x+2)}$

$$I = \int_0^3 \left(\frac{2x+3}{x^2+3} \right) \cdot dx - \int_0^3 \frac{2}{x+2} \cdot dx.$$

$$I = \left[\ln(x^2+3) \right]_0^3 + \left[\frac{3}{\sqrt{3}} \cdot \tan^{-1} \left(\frac{x}{\sqrt{3}} \right) \right]_0^3 - \left[2 \ln(x+2) \right]_0^3$$

$$I = [\ln 12 - \ln 3] + \sqrt{3} \tan^{-1} \sqrt{3} - 0 - 2 \ln 5 + 2 \ln 2.$$

$$I = \ln 4 + \sqrt{3} \cdot \frac{\pi}{3} - \ln 25 + \ln 4.$$

4

$$I = 2 \ln 4 - \ln 25 + \frac{\pi}{\sqrt{3}}$$

$$I = \ln \frac{16}{25} + \frac{\pi}{\sqrt{3}} \quad \text{OR} \quad \ln \left(\frac{16}{25} \right) + \frac{\sqrt{3}\pi}{3}$$

Q5(b) (i) $y = \frac{x(x-2)}{x-1}$

$x \neq 1$

$x=0, y=0$

$x \rightarrow \infty$

$x=2, y=0$

$y \rightarrow \infty$

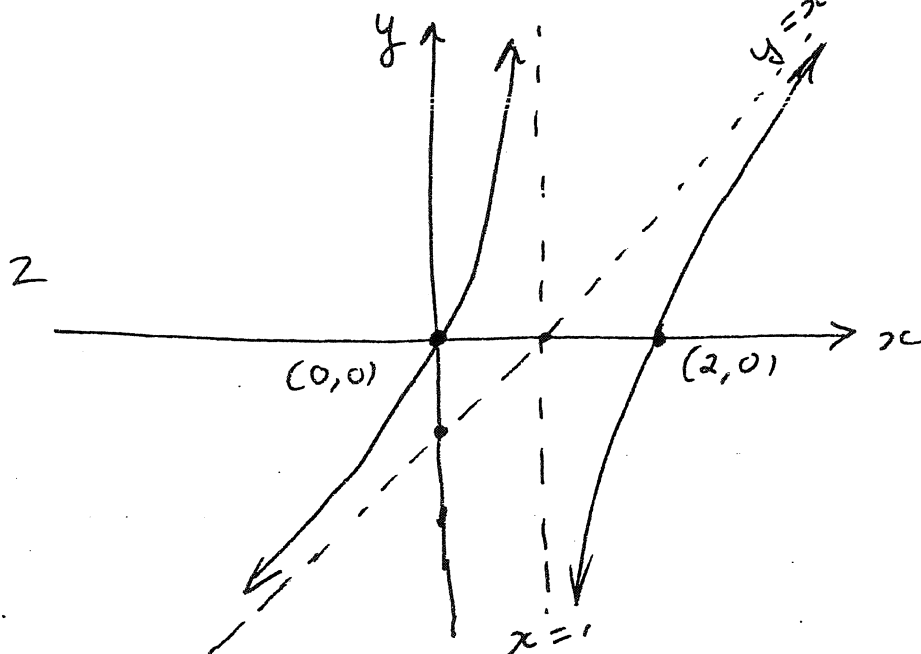
$x \rightarrow -\infty$

$y \rightarrow -\infty$

$$y = \frac{x^2 - 2x}{x-1} = \frac{x(x-1) - x}{x-1}$$

$$y = x - \frac{x}{x-1} \quad \text{as } x \rightarrow \infty$$

$$y = \underline{x-1}$$



(ii) $y = \frac{x(x-1)}{x-2}$

$x=0$

$x=1$

$x \neq 2$

$y=0$

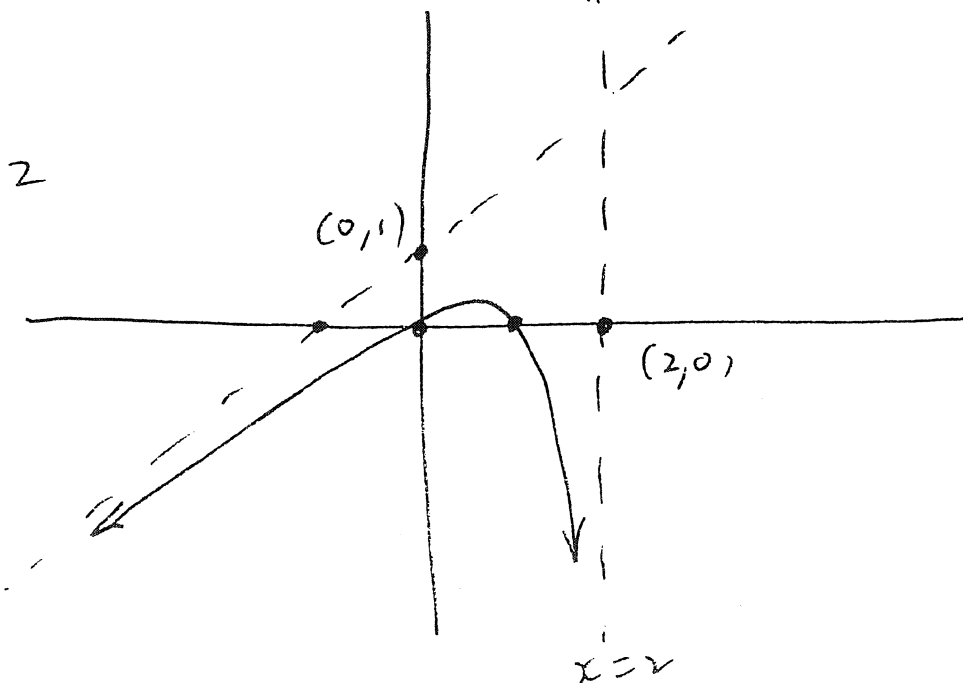
$y=0$

$$y = \frac{x^2 - x}{x-2}$$

$$y = \frac{x(x-2)}{x-2} + \frac{x}{x-2}$$

$$y = x + \frac{x}{x-2} \quad \text{as } x \rightarrow \infty$$

$$y = x+1$$



Q5(c). $y = \frac{x^3}{(x-1)^2}$

$x \neq 1$.

$x=0, y=0$

$y = x+2 + \frac{3x-2}{(x-1)^2}$

$x \rightarrow \infty$

$y = x+2$

(i) Asymptotes $\left. \begin{array}{l} x=1 \\ y=x+2 \end{array} \right\}$

$$\begin{array}{r} x+2 \\ x^2-2x+1 \overline{) x^3} \\ \underline{x^3-2x^2+x} \\ 2x^2-2x \\ \underline{2x^2-4x+2} \\ 3x-2 \end{array}$$

(ii) $\frac{dy}{dx} = \frac{3(x-1)^2 x^2 - x^3 \cdot 2(x-1)}{(x-1)^4}$
 $= \frac{3x^2(x-1) - 2x^3}{(x-1)^3} = \frac{3x^3 - 3x^2 - 2x^3}{(x-1)^3} = \frac{x^3 - 3x^2}{(x-1)^3}$

Put $\frac{dy}{dx} = 0$

$\therefore x^3 - 3x^2 = 0 \quad x^2(x-3) \Rightarrow x=0, x=3$

Test $\frac{dy}{dx}$ at $x=0$.

$x < 0 \quad \frac{dy}{dx} > 0$

$x > 0 \quad \frac{dy}{dx} > 0$

$\therefore$ horizontal inflexion at $(0,0)$

Test at $x=3$.

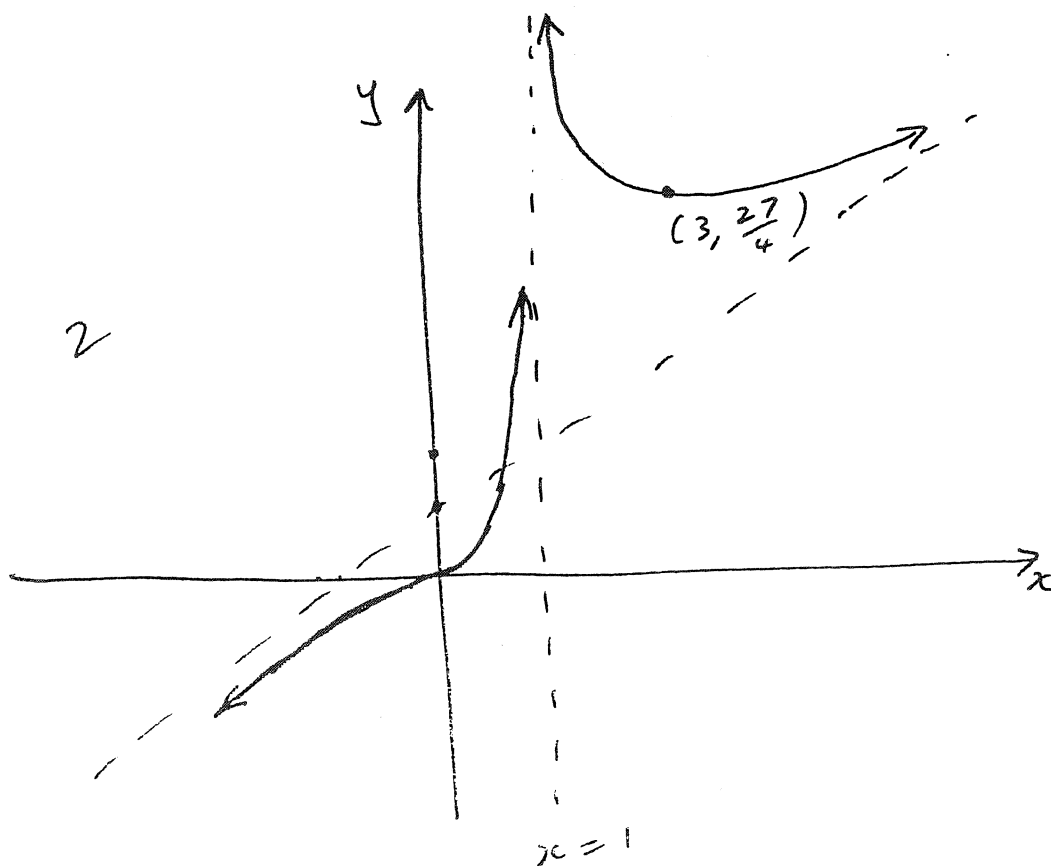
$x < 3$

$x > 3$

$\frac{dy}{dx} < 0$

$\frac{dy}{dx} > 0$

$\therefore$ MIN at $x=3$
 $y = \frac{27}{4}$



Q6 (a) i) $P(x) = (x-b)^m \cdot Q(x)$.

$$P'(x) = m(x-b)^{m-1} \cdot Q(x) + (x-b)^m \cdot Q'(x)$$

$$= (x-b)^{m-1} \{ m Q(x) + (x-b) \cdot Q'(x) \}$$

$$= (x-b)^{m-1} \cdot S(x)$$

$\therefore x=b$ is a root of multiplicity $m-1$ for $P'(x)$.

(ii) $Q(x) = x^4 - 5x^3 + 4x^2 + 3x + 9$.

$$Q'(x) = 4x^3 - 15x^2 + 8x + 3$$

$$Q'(1) = 0$$

$$Q(1) = 1 - 5 + 4 + 3 + 9 \neq 0$$

$\therefore x=1$ is not a zero.

$$Q'(3) = 4 \times 3^3 - 15 \times 3^2 + 8 \times 3 + 3$$

$$= 5 \times 27 - 9 \times 15 = 0$$

Try $Q(3) = 3^4 - 5 \times 3^3 + 4 \times 3^2 + 3 \times 3 + 9$

$$= 81 - 135 + 36 + 9 + 9$$

$$= 0$$

$\therefore x=3$ is a double root

$\therefore (x-3)^2$ is a factor

$$x^2 - 6x + 9$$

$$\begin{array}{r} x^2 + x + 1 \\ x^2 - 6x + 9 \overline{) x^4 - 5x^3 + 4x^2 + 3x + 9} \\ \underline{- x^4 - 6x^3 + 9x^2} \\ x^3 - 5x^2 + 3x \\ \underline{x^3 - 6x^2 + 9x} \\ x^2 - 6x + 9 \\ \underline{x^2 - 6x + 9} \\ 0 \end{array}$$

$$x^2 + x + 1$$

$$x = \frac{-1 \pm \sqrt{-3}}{2}$$

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Over the Real Field

$$\therefore x = 3, \frac{-1 \pm i\sqrt{3}}{2}$$

⑥ ⑥ α, β, γ . $3x^3 + 4x^2 + 5x + 1 = 0$

$$\frac{1}{\alpha^2 \beta^2} + \frac{1}{\beta^2 \gamma^2} + \frac{1}{\gamma^2 \alpha^2}$$

$$= \frac{\gamma^2 + \alpha^2 + \beta^2}{\alpha^2 \beta^2 \gamma^2}$$

$$\alpha + \beta + \gamma = -\frac{4}{3}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{5}{3}$$

$$\alpha\beta\gamma = -\frac{1}{3} \Rightarrow (\alpha\beta\gamma)^2 = \frac{1}{9}$$

$$(\alpha + \beta + \gamma)^2 = (\alpha + \beta + \gamma)(\alpha + \beta + \gamma)$$

$$= \alpha^2 + \alpha\beta + \alpha\gamma$$

$$+ \beta\alpha + \beta^2 + \beta\gamma + \gamma\alpha + \beta\gamma + \gamma^2$$

4

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$= \left(-\frac{4}{3}\right)^2 - 2\left(\frac{5}{3}\right)$$

$$= \frac{16}{9} - \frac{10}{3}$$

$$= \frac{16}{9} - \frac{30}{9} = -\frac{14}{9}$$

$$\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha^2 \beta^2 \gamma^2} = \frac{-\frac{14}{9}}{\frac{1}{9}} = \underline{\underline{-14}}$$

$$(c)(i) \quad x^2 - 3x + 4 = 0 \Rightarrow 9x^2 = 9(3x - 4)$$

$$x^2 = 3x - 4$$

$$x^4 = (3x - 4)^2 = 9x^2 - 24x + 16$$

$$= 9(3x - 4) - 24x + 16$$

$$= 27x - 36 - 24x + 16$$

$$\underline{x^4 = 3x - 20}$$

$$\alpha + \beta = 3$$

$$\alpha\beta = 4$$

2

(ii) Roots of $x^2 - 3x + 4 = 0$

α and β .

$$\alpha^4 = 3\alpha - 20$$

$$\alpha + \beta = 3$$

$$\beta^4 = 3\beta - 20$$

$$\alpha\beta = 4$$

$$\alpha^4 \beta^4 = (\alpha\beta)^4$$

$$= (4)^4 = \underline{256}$$

$$\alpha^4 + \beta^4 = 3(\alpha + \beta) - 40$$

$$\therefore \underline{\alpha^4 + \beta^4 = -31}$$

$$\cancel{\alpha^4} + \cancel{\beta^4} = (\alpha + \beta)^4$$

$$= \alpha^4 + 4\alpha^3\beta + 6\alpha^2\beta^2$$

$$+ 4\alpha\beta^3 + \beta^4$$

$$(\alpha\beta)^4 = \alpha^4\beta^4$$

$$= 4^4$$

$$= \underline{256}$$

$$\therefore \alpha^4 + \beta^4 = (\alpha + \beta)^4 - 4\alpha\beta(\alpha^2 + \beta^2)$$

$$= 6(\alpha^2\beta^2)$$

$$= (3)^4 - 4 \cdot 4(\alpha^2 + \beta^2) - 6(4)^2$$

3

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= 3^2 - 2 \cdot 4$$

$$= 1$$

$$= 3^4 - 16(1) - 6 \cdot (4)^2$$

$$= 81 - 16 - 96 = 81 - 112 = \underline{-31}$$

$$\therefore \underline{x^2 + 31x + 256 = 0}$$

Q6 (ii)

$$x^2 - 3x + 4 = 0$$

Roots α and β

$$\alpha + \beta = 3$$

$$\alpha\beta = 4.$$

$$x^4 = 3x - 20.$$

$$\therefore \alpha^4 = 3\alpha - 20$$

$$\alpha^4 + \beta^4 = 3(\alpha + \beta) - 2 \times 20$$

$$\beta^4 = 3\beta - 20.$$

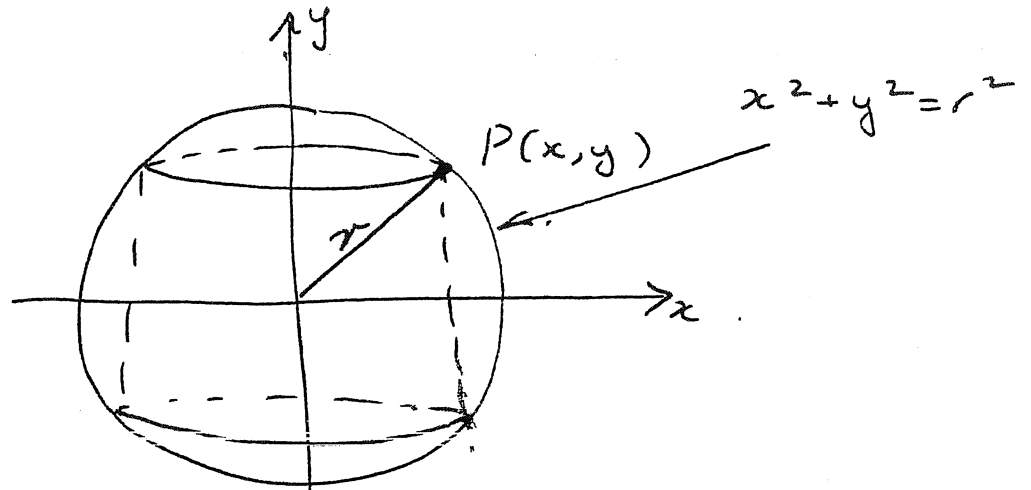
$$= 9 - 40 = -31.$$

$$\therefore \text{sum of roots} = -31.$$

$$\alpha^4 \beta^4 = (\alpha\beta)^4 = 4^4 = 256.$$

$$\therefore \text{Egn } X^2 + 31X + 256 = 0$$

Q7 @



Volume of a Cylinder $V = \pi R^2 H$.

From diagram $V = \pi (x^2) \times 2y$
 $= 2\pi x^2 y$

But $x^2 = r^2 - y^2 \Rightarrow V = 2\pi (r^2 - y^2) y$
 $= 2\pi y r^2 - 2\pi y^3$

For max volume require $\frac{dV}{dy}$

$$\frac{dV}{dy} = 2\pi r^2 - 6\pi y^2.$$

Put $\frac{dV}{dy} = 0 \Rightarrow \therefore r^2 = 3y^2 \Rightarrow y = \pm \frac{r}{\sqrt{3}}$

But y is a distance $\therefore y > 0$

$$\therefore y = \frac{r}{\sqrt{3}}.$$

$$\frac{d^2V}{dy^2} = -12\pi y < 0 \text{ for } y > 0$$

$\therefore$ Max Volume for $y = \frac{r}{\sqrt{3}}$

Max Volume $V = 2\pi y (r^2 - y^2) = 2\pi \frac{r}{\sqrt{3}} (r^2 - \frac{r^2}{3})$
 $= \frac{2\pi r}{\sqrt{3}} (\frac{3r^2 - r^2}{3})$

$$V = \frac{4\pi r^3}{3\sqrt{3}} \text{ c.u.}$$

(4)

Q7 (i) $z^5 = 1$ let $z = \text{cis } \theta$ $|z| = r = 1$

then $(\text{cis } \theta)^5 = \text{cis } (0 + 2k\pi)$ k an integer.

$\therefore \text{cis } 5\theta = \text{cis } (0 + 2k\pi)$ $k = 0, 1, 2, 3, 4$

By De Moivre's Theorem

$\therefore 5\theta = 0 + 2k\pi$

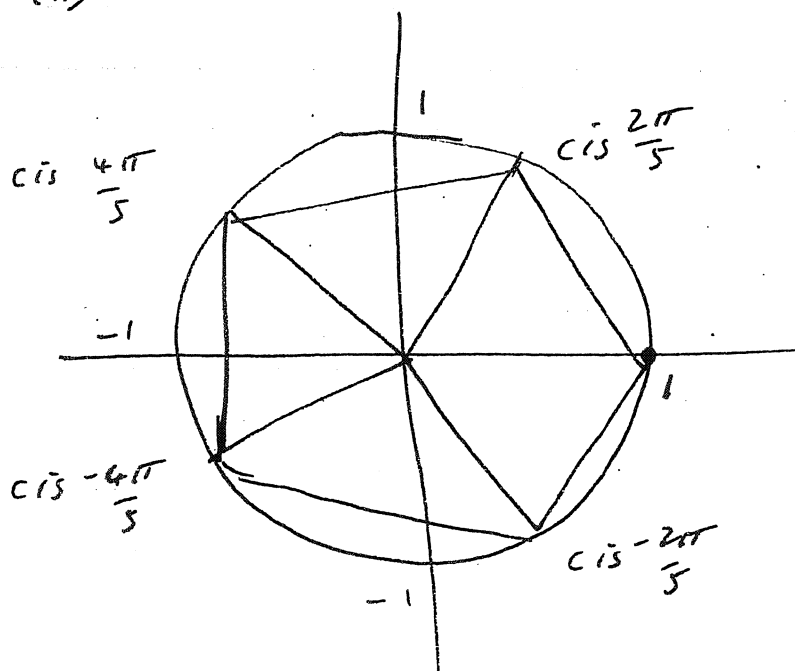
$\theta = \frac{2k\pi}{5}$ where $k = 0, 1, 2, 3, 4$.

$\therefore \theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5} (-\frac{4\pi}{5}), \frac{8\pi}{5} (-\frac{2\pi}{5})$

3

(3) $\therefore 5$ roots are $\text{cis } 0 = 1, \text{cis } \pm \frac{2\pi}{5}, \text{cis } \pm \frac{4\pi}{5}$

(ii)



Area of $\Delta = \frac{1}{2} ab \sin c = \frac{1}{2} \cdot 1 \cdot 1 \cdot \sin \frac{2\pi}{5}$.

Area of Pentagon $= 5 \times \frac{1}{2} \sin \frac{2\pi}{5}$

(2)

$= \frac{5}{2} \sin \frac{2\pi}{5}$ S.V.

2

Q7(b)(iii) $z^5 - 1 = (z - 1)(z - z_1)(z - z_2)(z - z_3)(z - z_4)$

But $z_1 = \text{cis } \frac{2\pi}{5}$ $z_2 = \text{cis } \frac{4\pi}{5}$
 $z_4 = \text{cis } -\frac{2\pi}{5}$ $z_3 = \text{cis } -\frac{4\pi}{5}$

(2)

with $z_1 = \bar{z}_4$ and $z_2 = \bar{z}_3$

2 $\therefore z^5 - 1 = (z - 1)(z^2 - 2\cos \frac{2\pi}{5}z + 1)(z^2 - 2\cos \frac{4\pi}{5}z + 1)$
 Roots in conjugate pairs.

(iv) Now $(z^5 - 1) = (z - 1)(z^4 + z^3 + z^2 + z + 1)$.

$\therefore z^4 + z^3 + z^2 + z + 1 = (z^2 - 2\cos \frac{2\pi}{5}z + 1)(z^2 - 2\cos \frac{4\pi}{5}z + 1)$

Equating coefficients of z^3
 $1 = -2\cos \frac{2\pi}{5} - 2\cos \frac{4\pi}{5}$

But. $\cos 2\theta = 2\cos^2 \theta - 1 \Rightarrow 2\cos \frac{4\pi}{5} = 2\{2\cos^2 \frac{2\pi}{5} - 1\}$

$\therefore 1 = -2\cos \frac{2\pi}{5} - 2\{2\cos^2 \frac{2\pi}{5} - 1\}$

$1 = -2\cos \frac{2\pi}{5} - 4\cos^2 \frac{2\pi}{5} + 2$

$\Rightarrow 4\cos^2 \frac{2\pi}{5} + 2\cos \frac{2\pi}{5} = 1$

Above eqn of form $4x^2 + 2x - 1 = 0$.

4 Exact Value: $x = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{-2 \pm \sqrt{20}}{8}$

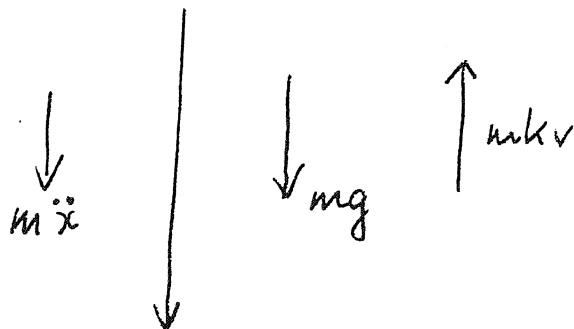
$x = \frac{-2 \pm 2\sqrt{5}}{8} = \frac{-1 \pm \sqrt{5}}{4}$

But $\frac{2\pi}{5} < \frac{\pi}{2} \Rightarrow \therefore \cos \frac{2\pi}{5} > 0$

(4)

$\therefore \cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}$

Q8 @



$$(i) \quad m\ddot{x} = mg - mkv$$

$$\therefore \ddot{x} = g - kv$$

$$\ddot{x} = \frac{v dv}{dx}$$

(1)

$$\therefore v \frac{dv}{dx} = g - kv$$

$$\text{at } t=0 \quad v=0 \quad x=0$$

$$v=V \quad x=h$$

$$(ii) \quad \frac{dx}{v \cdot dv} = \frac{1}{g - kv}$$

$$\int_0^h dx = \int_0^V \frac{v \cdot dv}{g - kv}$$

$$h = -\frac{1}{k} \int_0^V \frac{g - kv}{g - kv} \cdot dv + \frac{1}{k} \int_0^V \frac{g}{g - kv} \cdot dv$$

$$h = -\frac{V}{k} + -\frac{g}{k^2} \int_0^V \frac{-k \cdot dv}{g - kv}$$

$$h = -\frac{V}{k} - \frac{g}{k^2} \ln [g - kv]_0^V$$

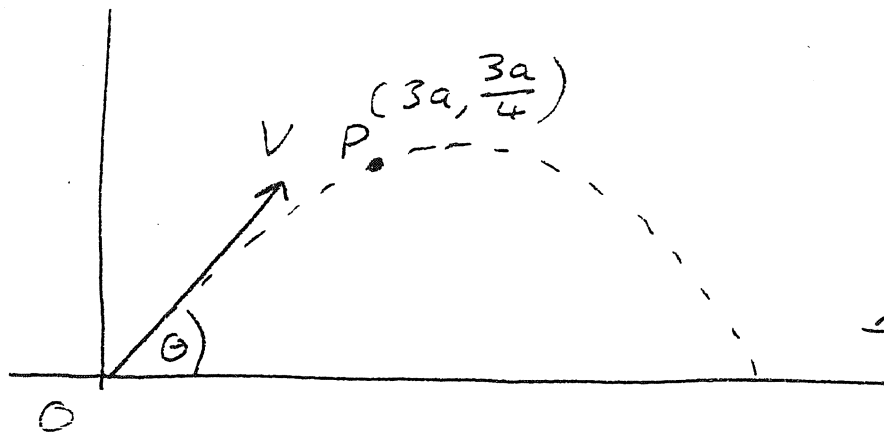
$$h = -\frac{V}{k} - \frac{g}{k^2} \ln \left[\frac{g - kv}{g} \right]$$

$$k^2 h = -Vk - g \ln \left[1 - \frac{kv}{g} \right]$$

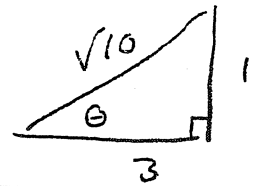
$$(4) \quad \therefore g \ln \left[1 - \frac{kv}{g} \right] + Vk + k^2 h = 0$$

$$\ln \left[1 - \frac{kv}{g} \right] + \frac{kv}{g} + \frac{k^2 h}{g} = 0$$

Q8 (b)(i)



$$\tan \theta = \frac{1}{3}$$



$$\sin \theta = \frac{1}{\sqrt{10}}$$

$$\cos \theta = \frac{3}{\sqrt{10}}$$

Vertically:

$$\ddot{y} = -g$$

$$\dot{y} = -gt + c_1$$

$$\text{at } t=0 \quad \dot{y} = \frac{V}{\sqrt{10}}$$

$$\therefore \dot{y} = -gt + \frac{V}{\sqrt{10}}$$

$$y = -\frac{1}{2}gt^2 + \frac{Vt}{\sqrt{10}} + c_2$$

$$\text{at } t=0 \quad y = 0$$

Horizontally

$$\ddot{x} = 0$$

$$\dot{x} = c_3$$

$$\text{at } t=0 \quad \dot{x} = \frac{3V}{\sqrt{10}}$$

$$\therefore \dot{x} = \frac{3V}{\sqrt{10}}$$

$$x = \frac{3Vt}{\sqrt{10}} + c_4$$

$$\text{at } t=0 \quad x = 0$$

$$\therefore y = -\frac{1}{2}gt^2 + \frac{Vt}{\sqrt{10}} \quad \text{--- (1)}$$

$$\therefore x = \frac{3Vt}{\sqrt{10}} \quad \text{--- (2)}$$

Now passes thro $(3a, \frac{3a}{4})$.

$$\text{In (2)} \quad 3a = \frac{3Vt}{\sqrt{10}} \Rightarrow t = \frac{a\sqrt{10}}{V}$$

$$\text{In (1)} \quad \frac{3a}{4} = -\frac{1}{2}gt^2 + \frac{Vt}{\sqrt{10}}$$

$$\frac{3a}{4} = -\frac{1}{2}g \left(\frac{a\sqrt{10}}{V} \right)^2 + \frac{V}{\sqrt{10}} \cdot \frac{a\sqrt{10}}{V}$$

$$\frac{3a}{4} = -\frac{1}{2}g \cdot \frac{a^2 10}{V^2} + a$$

$$\text{(5)} \quad \therefore \frac{1}{2}g \cdot \frac{a^2 10}{V^2} = \frac{a}{4}$$

$$2ag \cdot 10 = V^2$$

$$\therefore \underline{\underline{V^2 = 20ga}}$$

Q8(b) (ii)

For max height for Particle P $y = 0$

$$\therefore \frac{V}{\sqrt{10}} = gt \Rightarrow t = \frac{V}{g\sqrt{10}}$$

For time of flight for P $y = 0 \Rightarrow \frac{1}{2}gt^2 = \frac{Vt}{\sqrt{10}} \quad t=0$

$$t = \frac{2V}{g\sqrt{10}}$$

Range of flight $x = 3 \frac{Vt}{\sqrt{10}}$

$$x = \frac{3V}{\sqrt{10}} \cdot \frac{2V}{g\sqrt{10}} = \frac{3V^2}{5g}$$

For Q let u = velocity and angle α .

Then $y = -\frac{1}{2}gt^2 + ut \sin \alpha$ and $x = ut \cos \alpha$ (From (i),

When $t = \frac{V}{g\sqrt{10}} \quad y=0$ and $x = \frac{3V^2}{5g}$

$$\therefore \frac{1}{2}gt = u \sin \alpha$$

$$u \sin \alpha = \frac{V}{2\sqrt{10}} \quad \text{--- (3)}$$

$$\frac{3V^2}{5g} = u \cdot \frac{V}{g\sqrt{10}} \cos \alpha$$

$$u \cos \alpha = \frac{3\sqrt{10}V}{5} \quad \text{--- (4)}$$

$$\textcircled{3}^2 \Rightarrow u^2 \sin^2 \alpha = \frac{V^2}{40}$$

$$\textcircled{4}^2 \Rightarrow \frac{90}{25} V^2 = u^2 \cos^2 \alpha$$

$$\therefore u^2 (\sin^2 \alpha + \cos^2 \alpha) = \frac{V^2}{40} + \frac{90}{25} V^2 \Rightarrow \frac{29V^2}{8}$$

But $V^2 = 20ga$

$$\therefore u^2 = \frac{29 \times 20ga}{8} \Rightarrow u^2 = \frac{145ga}{2}$$

$$u = \sqrt{\frac{145ga}{2}} \quad *$$

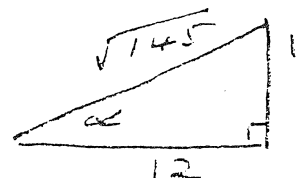
Now $V^2 = 40 u^2 \sin^2 \alpha$

$$20ga = 40 \cdot \frac{145ga}{2} \sin^2 \alpha$$

$$\therefore \sin^2 \alpha = \frac{1}{145} \Rightarrow \sin \alpha = \frac{1}{\sqrt{145}}$$

$$\therefore \tan \alpha = \frac{1}{12}$$

$$\alpha = \tan^{-1}\left(\frac{1}{12}\right)$$



①